Integer participation, structural noise, and discrete base frequencies in superconducting and superfluid clusters

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We revisit the "mother frequency" construction $F_{\mathrm{m}} = \Theta_{\mathrm{D}}/T_{\mathrm{c}}$ for a heterogeneous dataset of superconductors and superfluid ^{4}He, and ask whether the resulting scale behaves as a genuine discrete oscillator count once structural noise is taken into account. We build a phenomenological noise model on the same dataset, using structural topology and pressure descriptors to estimate how much each observed critical temperature T_{c} deviates from an idealised, noise–free value. This yields an idealised transition temperature $T_{\mathrm{c,\,ideal}\,i} = T_{\mathrm{c}\,i}(1 + \widehat{\xi}_i)$ and a noise–corrected mother frequency $F^*_{\mathrm{m}}i = \Theta_{\mathrm{D}}i/T_{\mathrm{c,\,ideal}}i$, where $\widehat{\xi}_i$ is the predicted structural noise for material i. For each family k we calibrate a base frequency $f_{\mathrm{base},k}$ that minimizes a robust loss between the integer participation $N_{\mathrm{corr}\,i} = F^*_{\mathrm{m}}i/f_{\mathrm{base},k}$ and the nearest integer. We then compare the distribution of distances $|\delta_i| = |N_{\mathrm{corr}\,i} - \mathrm{round}(N_{\mathrm{corr}\,i})|$ with null ensembles obtained by shuffling Θ_{D} or sampling from continuous surrogate distributions.

Across all families, the real data show a strong excess of near-integer participation relative to nulls: Kolmogorov–Smirnov and Mann–Whitney tests yield $p \simeq 4\times10^{-13}$, and Cliff's $\Delta \simeq 0.32$ indicates a medium effect size. At the family level, the median locking error $|\delta|$ varies systematically: molecular and oxide superconductors show tight integer locking, while high-pressure, binary and superfluid systems populate higher, noisier harmonics. Superfluid ^{4}He calibrates to $f_{\mathrm{base,He}} \simeq 4.0$, and the global median of $f_{\mathrm{base},k}$ across all families is $\tilde{f}_{\mathrm{base}} \simeq 4.16$, suggesting an approximate 1–2–4 ladder of preferred scales. Materials with the best integer locking reside in systematically quieter regions of the structural noise landscape (Cliff's $\Delta \simeq -0.47$ between the top 20% most integer-like and the rest), and the absolute locking error $|\delta|$ grows with the family–normalized noise Z_ξ. Overall, the results support a coarse picture in which a small number of discrete base frequencies, modulated by structural noise, organize both superconducting and superfluid transition scales into an integer participation pattern. We stress that this structure is phenomenological and approximate: whether it reflects a deeper oscillator-based mechanism or hidden regularities in curated experimental data remains an open question.

I. INTRODUCTION

Conventional descriptions of superconductivity and superfluidity relate the critical temperature T_{c} and characteristic excitation energies to microscopic parameters such as electron–phonon coupling, density of states, and pairing interactions [? ?]. In previous DOFT-inspired work we explored a complementary, highly compressed view, in which transition scales are organised by a "mother frequency" $F_{\mathrm{m}} = \Theta_{\mathrm{D}}/T_{\mathrm{c}}$ and a discrete locking grammar reminiscent of delayed oscillator networks with memory kernels [? ?]. The surprising outcome was that a heterogeneous set of superconductors and superfluid helium could be mapped into robust "prime-space fingerprints", with relatively small residuals after a simple two-parameter correction.

That earlier analysis treated structural noise implicitly, through a coarse correction law. Here we make the noise layer explicit and ask a sharper question: does the mother frequency behave as an integer participation count once we correct for a phenomenological estimate of structural and pressure-induced distortions? To address this we build, on the same dataset, a regression model that predicts how much each observed T_{c} deviates from an idealised, noise-free value based on simple structural and pressure descriptors. We then treat that model as a phenomenological layer and study the integer structure of the resulting noise-corrected mother frequencies.

The perspective adopted here is deliberately modest. We do not attempt to derive F_{m} or its discreteness from a microscopic Hamiltonian. Instead, we ask whether a simple integer–participation picture has empirical support once the main sources of noise are factored out, and whether the resulting patterns correlate with independently estimated structural noise. The outcome will be interpreted in the general spirit of mode–locking phenomena in driven oscillator systems [? ?]: integer and rational ratios as empirically favoured organising scales, not as exact microscopic invariants.

II. DATA AND STRUCTURAL NOISE

A. Materials and families

We work with an updated version of the curated dataset introduced in the previous DOFT-inspired study. Each row corresponds to a material and sub–network, labelled by a name and a category: SC_TypeI,

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SC_TypeII, SC_Binary, SC_Oxide, SC_Molecular, SC_HighPressure, SC_IronBased, SC_HeavyFermion, and a superfluid category for liquid ^{4}He. For each entry we record the critical temperature T_c, Debye temperature Θ_D, Fermi energy E_F where available, and qualitative structural information (dimensionality, layering, coordination, pressure range, etc.). This structural information feeds into the noise model described below.

B. Components of structural noise

The DOFT-inspired viewpoint suggests that three ingredients dominate the deviation between the observed critical temperature and a hypothetical "clean" value:

1. *Thermal coupling noise* associated with the jump from gap to Debye scales, which can smear the effective location of the transition relative to a sharp oscillator threshold.
2. *Structural noise* associated with cluster topology, inhomogeneity, and defects, which broadens and distorts the effective coupling landscape.
3. *Pressure noise* associated with large changes in external pressure along an otherwise similar structural path, particularly in high-pressure hydrides and related systems.

In practice these contributions are difficult to separate cleanly from published data. We therefore collapse them into a single effective noise amplitude $\widehat{\xi}_i$ for each material i, to be used as a phenomenological correction factor.

C. Phenomenological noise model

For each material we build a vector of descriptors x_i encoding:

- categorical family labels (Type I/II, binary, oxide, molecular, high-pressure, iron-based, heavy-fermion, superfluid);
- coarse structural features (layered vs three-dimensional, presence of light atoms, multi-band character);
- pressure information (ambient, moderate, extreme pressure, approximate pressure range where available);
- simple scale ratios such as Θ_D/T_c and gap-to-Debye ratios from the previous analysis.

We then fit a regression model of the form

$$\widehat{\xi}_i = f_\theta(x_i), \tag{1}$$

where f_θ is a regularised nonlinear regressor (gradient-boosted trees in the implementation used here), trained to minimise the mean absolute deviation between the observed T_{ci} and a leave-one-out estimate of a local family baseline. Hyperparameters are selected by cross-validation within the dataset, and performance is summarised by standard metrics (mean absolute error, R^2) as well as diagnostic plots provided in the Supplementary Material.

The resulting $\widehat{\xi}_i$ should not be interpreted as a microscopically precise decomposition of thermal, structural, and pressure effects. It is a coarse indicator of how much the observed T_{ci} deviates from an idealised value given the structural context of each material. We use it as a phenomenological knob to define a noise-corrected mother frequency.

D. Idealized T_c and mother frequency

Given $\widehat{\xi}_i$ we define an "idealised" critical temperature that would be observed in the absence of the inferred structural noise,

$$T_{c,\,\mathrm{ideal}\,i} = T_{ci}\big(1 + \widehat{\xi}_i\big), \tag{2}$$

and a corresponding noise–corrected mother frequency

$$F^*_{\mathrm{m}i} = \frac{\Theta_{\mathrm{D}i}}{T_{c,\,\mathrm{ideal}\,i}}. \tag{3}$$

For control we also retain the raw mother frequency $F_{\mathrm{m}i} = \Theta_{\mathrm{D}i}/T_{ci}$, and track both through the analysis.

Because different families live in very different structural regimes, the raw noise $\widehat{\xi}_i$ is not directly comparable between, say, high–pressure hydrides and low–temperature Type–I metals. To compare noise across the entire dataset we therefore define a family–normalized Z–score,

$$Z_{\xi i} = \frac{\widehat{\xi}_i - \mu_{\xi,k}}{\sigma_{\xi,k}}, \tag{4}$$

where $\mu_{\xi,k}$ and $\sigma_{\xi,k}$ are the mean and standard deviation of the predicted noise within family k. This rescaling allows us to ask whether materials that sit closer to integer participation counts tend to live in systematically quieter structural environments.

III. INTEGER PARTICIPATION MODEL

A. Base frequency calibration

For each family k we seek a base frequency $f_{\mathrm{base},k}$ such that the corrected mother frequencies $F^*_{\mathrm{m}i}$ in that family are well described by integer multiples,

$$N_{\mathrm{corr}\,i} = \frac{F^*_{\mathrm{m}i}}{f_{\mathrm{base},k(i)}}, \tag{5}$$

with $k(i)$ denoting the family of material i. We define the locking error for each material as the distance to the nearest integer,

$$\delta_i = \left| N_{\mathrm{corr}\,i} - \mathrm{round}(N_{\mathrm{corr}\,i}) \right|. \tag{6}$$

To calibrate $f_{\mathrm{base},k}$ we minimize a robust loss over materials i in family k,

$$\mathcal{L}_k(f_{\mathrm{base},k}) = \mathrm{median}_i\, |\delta_i| + \lambda\, \mathrm{mean}_i\, N_{\mathrm{corr}\,i}, \tag{7}$$

where the small regularisation term λ prevents the optimum from drifting to very high harmonics with artificially small residuals. We constrain the search to a physically reasonable range $0.5 \le f_{\mathrm{base},k} \le 5.0$, so that the resulting participation counts N_{corr} occupy a moderate range (typically $1 \lesssim N_{\mathrm{corr}} \lesssim 40$) and can be meaningfully compared across families.

We also explore a global–frequency variant in which a single f_{base} is used for all families. The qualitative conclusions are similar; here we focus on the family–wise calibration, which makes the contrast between families more transparent.

B. Null models and statistical tests

To test whether the observed locking structure could arise by chance, we construct null ensembles that preserve only coarse aspects of the data.

Shuffle null. We randomly permute the Debye temperatures $\Theta_{\mathrm{D}i}$ across materials, recompute $F^*_{\mathrm{m}\,\pi(i)} = \Theta_{\mathrm{D}\,\pi(i)}/T_{\mathrm{c,\,ideal}\,i}$ with the same noise model, recalibrate $f_{\mathrm{base},k}$, and recompute the locking errors $|\delta_i|$. This preserves the distribution of T_{c}, Θ_{D} and noise within each family but destroys any specific pairing between them.

Continuous null. As a more aggressive null we fit a simple continuous distribution (gamma or lognormal) to the corrected mother frequencies $F^*_{\mathrm{m}\,i}$ in each family, sample synthetic values from that distribution, and repeat the calibration and locking procedure. This removes even the discrete structure of the empirical $F^*_{\mathrm{m}\,i}$.

For both nulls we generate many realisations and compare the distribution of locking errors $|\delta_i|$ with the real data, using Kolmogorov–Smirnov and Mann–Whitney tests, and Cliff's Δ as a measure of effect size. Uncertainties and confidence intervals are estimated by nonparametric bootstrap resampling [?], and effect sizes are reported using Cliff's dominance statistic Δ [?], which is robust to non–Gaussian tails and directly interpretable as the probability that a randomly chosen sample from one group exceeds a randomly chosen sample from the other.

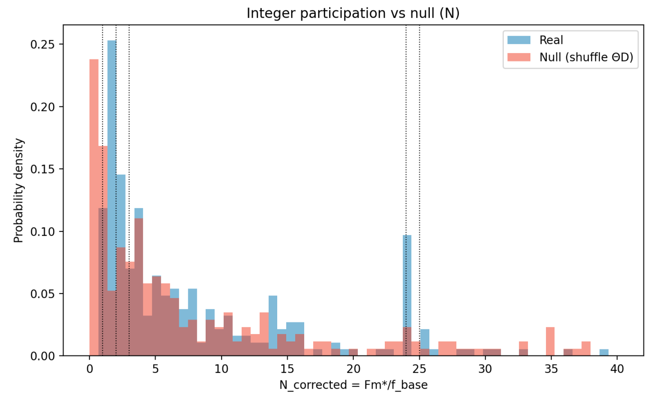

FIG. 1. Distribution of integer participation counts $N_{\mathrm{corr}} = F^*_{\mathrm{m}}/f_{\mathrm{base},k}$ in the real data (blue) compared with the shuffle null (orange). Both exhibit broad support up to $N_{\mathrm{corr}} \sim 40$, with clusters near low integers and around $N_{\mathrm{corr}} \simeq 24$, corresponding to families whose base frequency lies near unity.

IV. RESULTS

A. Integer participation vs null ensembles

Figure 1 compares the distribution of integer participation counts N_{corr} between the real data and the shuffle null. Both show a broad range of participation, from ~ 1 up to ~ 40, with clusters around low integers and around $N_{\mathrm{corr}} \simeq 24$. The latter corresponds to families whose base frequency sits closer to ~ 1 rather than ~ 4.

Figure 2 focuses on the locking error $|\delta|$. On a log–scaled density axis, the real data exhibit a clear excess of near–integer events compared to the shuffled null. Kolmogorov–Smirnov and Mann–Whitney tests yield $p \simeq 3.7 \times 10^{-13}$ and $p \simeq 2.7 \times 10^{-13}$ respectively, and Cliff's $\Delta \simeq 0.32$ indicates a medium effect size. In other words, the dataset contains substantially more near–integer mother frequencies (after noise correction) than would be expected from a generic reshuffling of Debye and critical temperatures.

B. Per–family locking strength

Figure 3 summarises the locking error $|\delta|$ by family. To avoid a few extreme outliers we clip $|\delta|$ at unity. The median error (orange line in each box) varies systematically between families: molecular and oxide superconductors show the smallest typical errors, followed by iron–based and heavy fermion systems; Type–I, Type–II, binary, high–pressure and superfluid families show broader, higher–median distributions, consistent with stronger distortions of the underlying oscillator network.

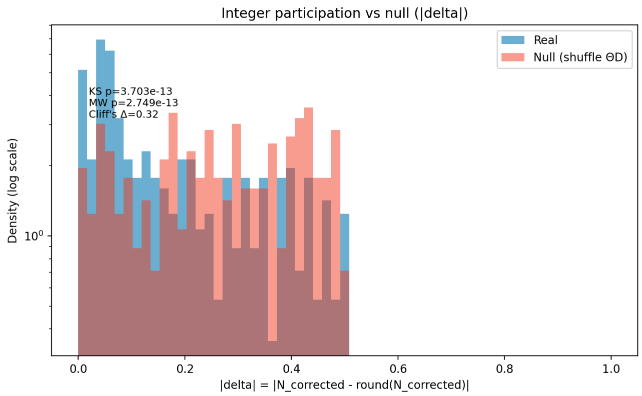

FIG. 2. Locking error $|\delta| = |N_{\rm corr} - \mathrm{round}(N_{\rm corr})|$ for the real data (blue) and shuffle null (orange), on a log–density scale. The real dataset shows a strong excess of near–integer events relative to the null. The overlaid annotation corresponds to Kolmogorov–Smirnov and Mann–Whitney p–values and Cliff's Δ between the two distributions.

C. Base frequencies across families

The calibrated base frequencies $f_{{\rm base},k}$ are shown in Figure 4, ordered by their median values. Superfluid helium sits at $f_{\rm base,He} \simeq 4.0$, while the global median across families is $\tilde{f}_{\rm base} \simeq 4.16$ (dashed line). Binary, molecular, high–pressure and Type–I superconductors cluster around this "four–like" scale. Oxides and heavy fermions sit near ~ 2.3, suggestive of an approximate half–harmonic of the same ladder, and iron–based systems calibrate near unity, corresponding to participation counts $N_{\rm corr}$ clustered around ~ 24 rather than ~ 6.

We do not interpret these values as exact rational ratios. Instead, we view them as approximate attractors within the allowed search range, consistent with a coarse 1–2–4 structure in the effective base frequencies.

D. Locking vs structural noise

We now relate the locking error $|\delta|$ to the family–normalized noise Z_ξ defined in Sec. II D. Figure 5 shows $|\delta|$ vs. Z_ξ for the per–family medians, with colour coding by family. There is a positive Spearman rank correlation $\rho \simeq 0.34$ (annotated in the plot), indicating that families with higher normalized noise tend to show weaker integer locking.

To probe this more directly we split the dataset into a "near–integer" group (materials in the lowest 20% of $|\delta|$) and the rest, and compare their noise distributions. Figure 6 shows that the near–integer group is systematically quieter: Cliff's $\Delta \simeq -0.47$ indicates that a randomly chosen near–integer material is substantially more likely to have lower noise than a randomly chosen member of the rest group.

V. DISCUSSION

A. A phenomenological 1–2–4 ladder

The most striking numerical feature of this analysis is the emergence of a small set of preferred base frequencies. Superfluid ^{4}He calibrates very close to $f_{\rm base,He} \simeq 4.0$, and the global median across superconducting families is $\tilde{f}_{\rm base} \simeq 4.16$. Several families—binary, Type–I, molecular and high–pressure superconductors—cluster around this scale, while oxides and heavy fermions sit near ~ 2.3, and iron–based superconductors near unity (Fig. 4).

The appearance of approximate 1–2–4 bands in the calibrated base frequencies is reminiscent of mode–locking ladders in driven and coupled oscillator systems, where rational ratios between effective frequencies organise the dynamics into so–called Arnold tongues [? ?]. Our construction is much more phenomenological: we do not derive the locking ratios from a specific microscopic Hamiltonian, and the integer participation counts $N_{\rm corr}$ are defined only at a coarse, family–dependent level. Nevertheless, the fact that a single noise–corrected mother frequency can be compressed into a small set of near–integer bands across superconductors and superfluid helium suggests that a similar mode–locking intuition may be a useful organising metaphor.

We do not interpret the 1–2–4 structure as a derived microscopic constant. The bands are broad, the ratios are only approximate, and the calibration procedure itself may bias the optimum towards moderate values of $f_{\rm base}$. From a conservative standpoint, the main statement is simply that a small number of effective scales seem sufficient to organise the transition temperatures of a heterogeneous dataset once structural noise is accounted for.

B. Role of structural noise

The correlations between integer locking and structural noise (Figs. 5 and 6) support a simple physical narrative. In this view, the mother frequency $F_{\rm m}$ measures the participation of a coarse oscillator network in the transition, but the ability to resolve that participation as an integer depends on the structural environment. In quiet, well–ordered systems (small Z_ξ), integer multiples of a base frequency can be resolved clearly, leading to small $|\delta|$. In noisy, strongly distorted environments, the same underlying oscillator structure may be present but is blurred: participation counts are still discrete in principle, but experimentally appear as smeared non–integers.

This narrative is consistent with the observed trends: families with strong structural distortions (high pressure, complex multiphase structures) show broader locking errors, while simpler molecular and oxide systems show tighter locking. It also explains why superfluid helium, despite its strong quantum coherence, sits in a relatively noisy band: the extracted structural noise includes the

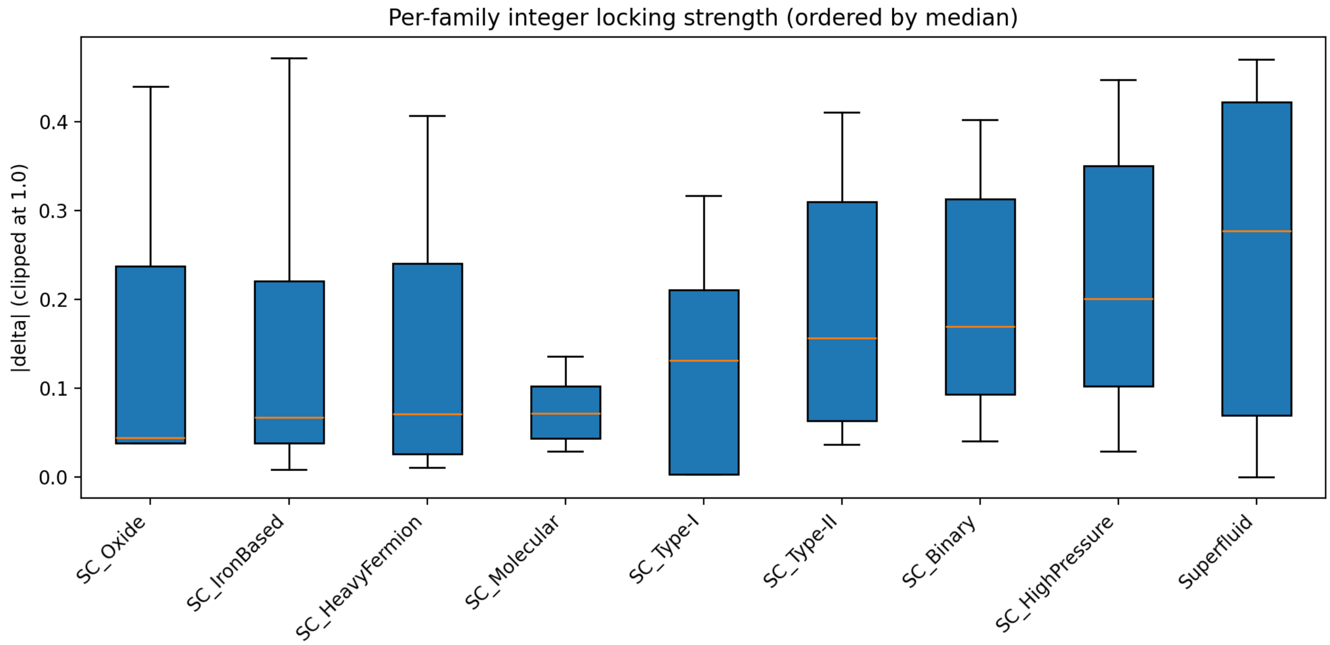

FIG. 3. Per–family distribution of locking errors $|\delta|$, clipped at 1.0 for visual clarity. Boxes show interquartile ranges, whiskers extend to 1.5 times the interquartile range, and outliers are plotted individually. Families are ordered by increasing median error. Molecular and oxide superconductors show the tightest integer participation, while high–pressure, binary and superfluid systems populate broader, noisier harmonics.

effect of large pressure ranges and the sensitivity of the roton mode itself.

C. Limitations and outlook

Several limitations of the present work are worth emphasising. First, the noise model is purely phenomenological and built from a limited set of structural descriptors. A more microscopic treatment, e.g. combining *ab initio* phonon calculations with realistic disorder, would be needed to anchor $\widehat{\xi}$ in specific mechanisms. Second, the integer participation picture has so far been tested only on a curated dataset; out–of–sample validation on new materials will be crucial. Third, we have not connected the base frequencies $f_{\mathrm{base},k}$ to concrete microscopic quantities such as characteristic phonon branches or collective modes. Doing so would require a bridge between the coarse mother frequency picture and the detailed excitation spectra of each family.

Despite these caveats, the combination of strong statistical departures from null ensembles, the systematic family structure of locking errors and base frequencies, and the correlation with structural noise suggests that the integer participation picture captures a nontrivial regularity of the superconducting and superfluid landscape. Whether this ultimately points to a deeper oscillator–based description, or simply exposes hidden regularities in curated data, is a question that future experiments and larger datasets will decide.

VI. CONCLUSIONS

We have shown that, after correcting for a phenomenological model of structural noise, the mother frequency $F_{\mathrm{m}} = \Theta_{\mathrm{D}}/T_{\mathrm{c}}$ behaves as an approximate integer participation count of family–dependent base frequencies in a heterogeneous set of superconductors and superfluid helium. The distribution of distances to the nearest integer is incompatible with simple shuffle and continuous null models, with p–values $\sim 10^{-13}$ and a medium effect size. Different families exhibit distinct locking strengths and base frequencies, with a suggestive 1–2–4 ladder connecting iron–based, oxide/heavy–fermion, and helium/binary/Type–I systems. Materials with the tightest integer locking occupy systematically quieter regions of structural noise.

We emphasise that the construction is phenomenological and deliberately conservative in its claims. The results do not establish a new microscopic theory of superconductivity or superfluidity. They do, however, show that a simple integer–participation grammar, coupled to a structural noise model, can organise a broad class of transition temperatures into a small number of discrete scales. This suggests new ways of summarising and probing complex materials datasets, and motivates further work on the connection between discrete oscillator net-

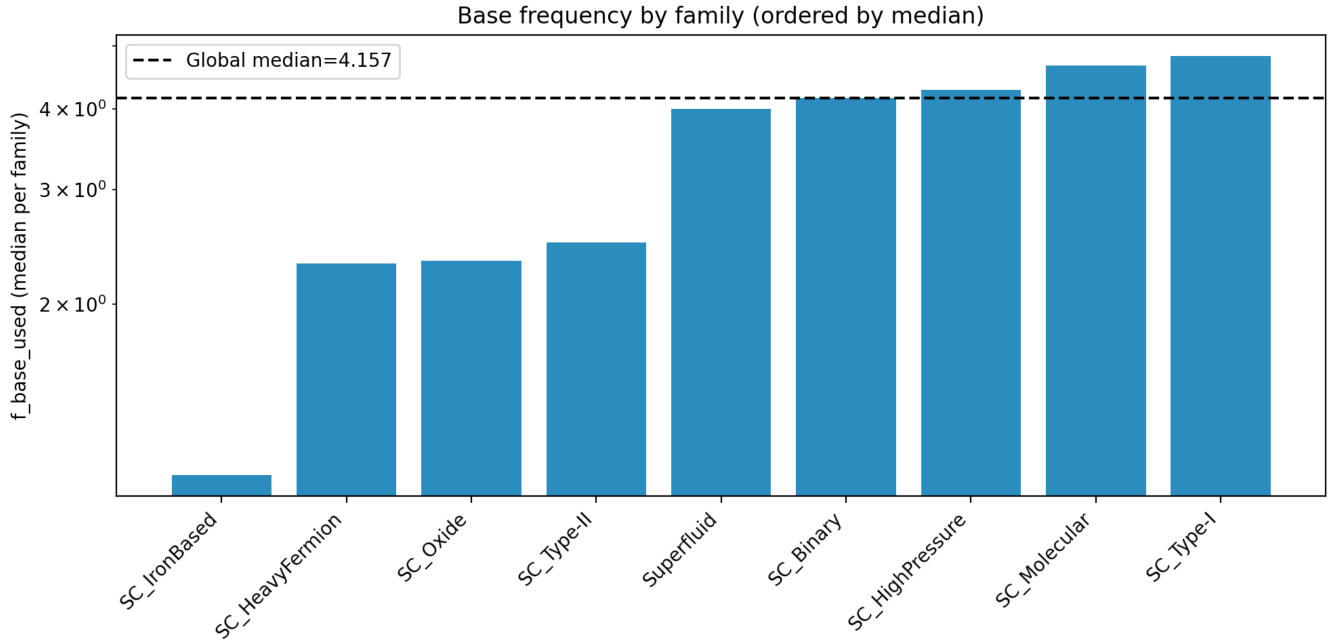

FIG. 4. Calibrated base frequencies $f_{\mathrm{base},k}$ (medians per family) ordered from left to right. The dashed line marks the global median $\tilde{f}_{\mathrm{base}} \simeq 4.16$. Superfluid helium calibrates near $f_{\mathrm{base,He}} \simeq 4.0$; several superconducting families cluster around the same scale, while oxides and heavy fermions sit near ~ 2.3 and iron–based materials near unity. These bands are consistent with an approximate 1–2–4 ladder of preferred base frequencies, though the structure is phenomenological and not exact.

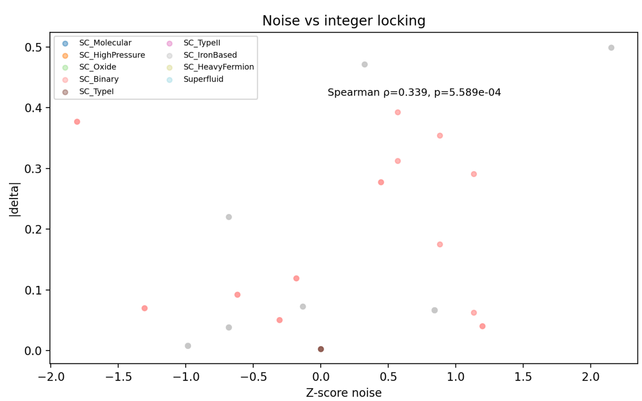

FIG. 5. Locking error $|\delta|$ vs. family–normalized structural noise Z_ξ, with points coloured by superconducting family. The annotation reports the Spearman rank correlation between the two quantities. Families with larger normalized noise tend to exhibit weaker integer participation.

works, structural noise, and emergent coherence in quantum matter.

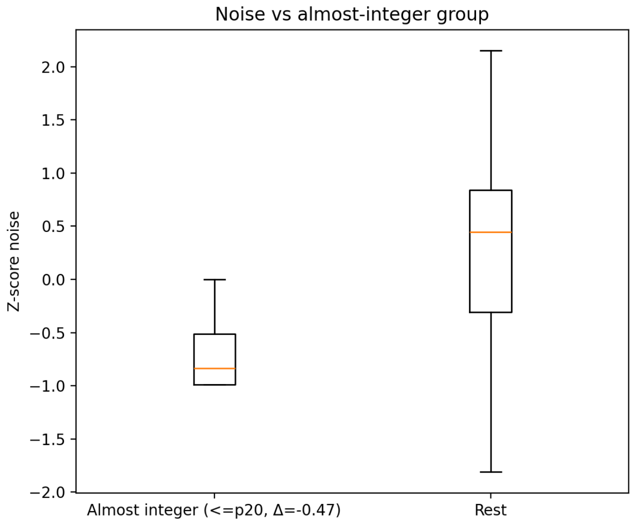

FIG. 6. Structural noise Z_ξ for two groups: materials in the lowest 20% of locking error ("almost integer") and the rest of the dataset. The almost–integer group resides in systematically quieter structural environments; Cliff's $\Delta \simeq -0.47$ quantifies the effect size.